$$\int x \sin \frac{x}{2} \, dx$$

$$u = x \Rightarrow du = dx$$
$$dv = \operatorname{sen}\left(\frac{x}{2}\right) dx \overset{(*)}{\Rightarrow} v = -2\cos\left(\frac{x}{2}\right)$$

(*) Precisamos fazer uma substituição

$$z = \frac{x}{2} \Rightarrow dz = \frac{1}{2} dx$$

$$\int \sin\left(\frac{x}{2}\right) dx = \int \sin(z)\, 2\, dz =$$
$$= 2\int \sin z \, dz = -2\cos\left(\frac{x}{2}\right) + k$$

Logo:

$$\int x \sin \frac{x}{2} \, dx = uv - \int v \, du =$$
$$= -x\, 2\cos\left(\frac{x}{2}\right) - \int -2\cos\left(\frac{x}{2}\right) dx =$$
$$= -2x\cos\left(\frac{x}{2}\right) + 2\int \cos\left(\frac{x}{2}\right) dx \overset{**}{=}$$
$$= -2x\cos\left(\frac{x}{2}\right) + 2\left(2\sin\frac{x}{2}\right) + K$$
$$= -2x\cos\left(\frac{x}{2}\right) + 4\operatorname{sen}\left(\frac{x}{2}\right) + K$$

$$\therefore \int x \sin \frac{x}{2} \, dx = -2x\cos\left(\frac{x}{2}\right) + 4\sin\left(\frac{x}{2}\right) + K$$

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** Resolvendo por substituição: $\int \cos\left(\frac{x}{2}\right) dx$

seja $z = \frac{x}{2} \Rightarrow dz = \frac{1}{2} dx$

daí

$$\int \cos\left(\frac{x}{2}\right) dx = \int \cos z \cdot 2\, dz =$$

$$= 2 \int \cos z\, dz = 2 \operatorname{sen}\left(\frac{x}{2}\right) + K$$

$$\int t^2 \cos t\, dt$$

$u = t^2 \Rightarrow du = 2t\, dt$

$dv = \cos t\, dt \Rightarrow v = \operatorname{sen} t$

$$\int t^2 \cos t\, dt = t^2 \operatorname{sen} t - \int \operatorname{sen}(t)\, 2t\, dt$$

$$= t^2 \operatorname{sen} t - 2 \int \operatorname{sen}(t)\, t\, dt \qquad (*)$$

Temos que resolver: $\int \operatorname{sen}(t)\, t\, dt$

$u = t \Rightarrow du = dt$

$dv = \operatorname{sen}(t)\, dt \Rightarrow v = -\cos(t)$

$$\int \operatorname{sen}(t)\, dt = uv - \int v\, du =$$

$$= -t \cos t - \int -\cos t\, dt =$$

$$= -t \cos t + \operatorname{sen}(t) + K$$

Voltando em (*):

$$\int t^2 \cos t \, dt = t^2 \operatorname{sen} t - 2\int \operatorname{sen}(t)\, t \, dt$$

$$= t^2 \operatorname{sen}(t) + 2t\cos(t) - 2\operatorname{sen}(t) + K$$

$$\int_0^{\pi} x \cos x^2 \, dx$$

$$u = x^2 \Rightarrow du = 2x\,dx \qquad \begin{cases} g(0) = 0 \\ g(\pi) = \pi^2 \end{cases}$$

$$\int_0^{\pi} x \cos x^2 \, dx = \int_0^{\pi^2} \cos(u) \frac{du}{2} \Rightarrow$$

$$\frac{1}{2}\int_0^{\pi^2} \cos u \, du = \frac{1}{2}\left(\operatorname{sen} u\right]_0^{\pi^2} =$$

$$= \frac{1}{2}\left[\operatorname{sen}(\pi^2) - \operatorname{sen} 0\right] = \frac{1}{2}\operatorname{sen}(\pi^2)$$

$$\int_{13}^{14} (x-13)^{10} dx \overset{(*)}{=} \int_0^1 u^{10} du =$$

$$= \left(\frac{u^{11}}{11}\right]_0^1 = \frac{1}{11} - \frac{0}{11} = \frac{1}{11}$$

(*) Seja $u = x - 13$ e $g(13) = 0$
$du = dx$ $\quad g(14) = 1$

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